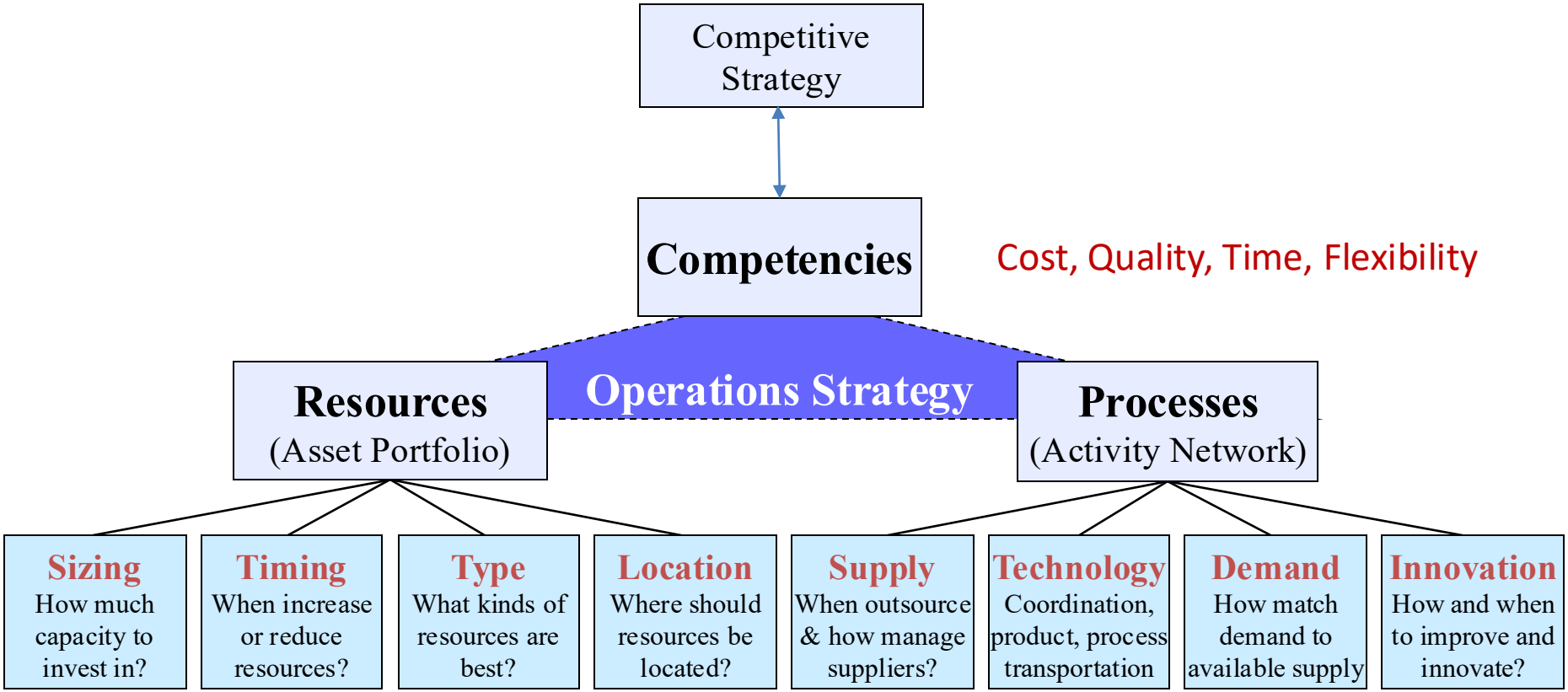
Case Analysis with our Framework



‘Design and manufacturing solution shop’

Competences/Capabilities view of operations

Flexibility (not only design, but also volume, schedule, and packaging) at reasonable costs

ACC
Operations Strategy

Resource View of operations

Size: capacity is 600M with 85% planned utilization rate, real utilization is only 70% (FR=420M)

Timing: no plan to change capacity, historically, its been leading in its capacity strategy

Type: generalized batch production in 5 separate functional areas

Location: USA and Europe

Process View of operations

Supply: 4500 SKUs, also emphasize variety in packaging, high WIP, low finished product inventory, quality inspection (26000 defects per million), indirect labour is high 56%, reduced suppliers base

Technology: focus on product technology and product design, process technology depends on suppliers (used to always have latest process technology, but lack of updates since 1988)

Innovation: incremental process technology updates (last update was 1988) and customized product design

Demand: make to order, sales controls production/engineering, weekly (at most 1.5 to 2 days) delivery, long lead time (10 days) for standard products

‘The leaner enterprise’

Competences/Capabilities view of operations

Very Low Cost and defects, on time delivery (sometimes many times a day)

DJC
Operations Strategy

Resource View of operations

Size: 800M capacity with planned 100% utilization, real utilization is 87.5% (FR=700M)

Timing: having been expanding and now plan to expand to USA

Type: dedicated/specilized 4 cells each has 4-6 repetitive lines, standardized packaging

Location: Japan and close to suppliers and customers, planning expanding to the USA

Process View of operations

Supply: 640 SKUs, standardized packaging, low variety, strong relationships with suppliers, indirect labour is low, low WIP, good quality through prevention (1 defects per million), 99% yield, long production runs, young skilled staff

Technology: focus on in house self-development process technology improvement and design

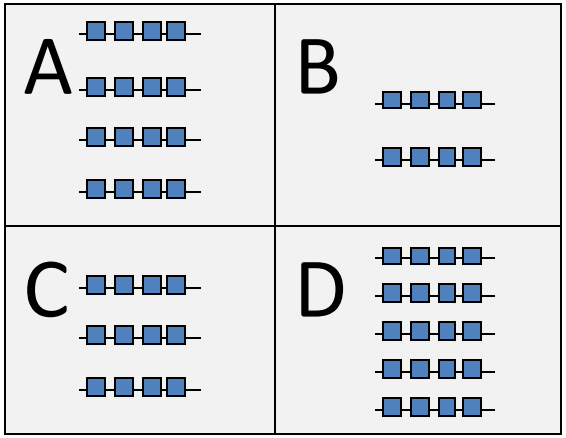
Innovation: reverse engineering, simplicity and manufacturability, pre-automation, continuous improvement, , JIT delivery

Demand: standardized design supplying 640 SKUs, close to customers, value engineering/production over sales

Accessing Alignment

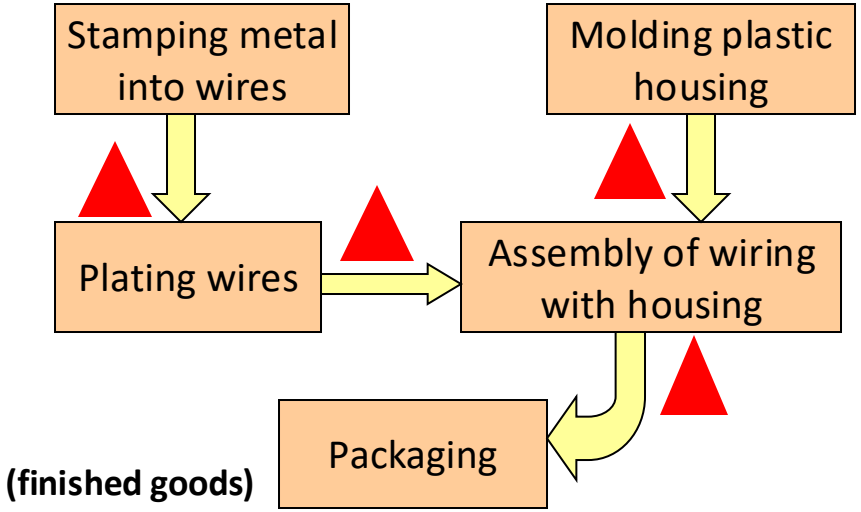
- Layout/process comparison of DJC and ACC

DJC: Connected line flows by Product family



- Plant divided into four cells
- Each cell has 2 to 6 parallel lines
- Each line has its own stamping, molding, assembly & packaging
- Plating is common to four cells

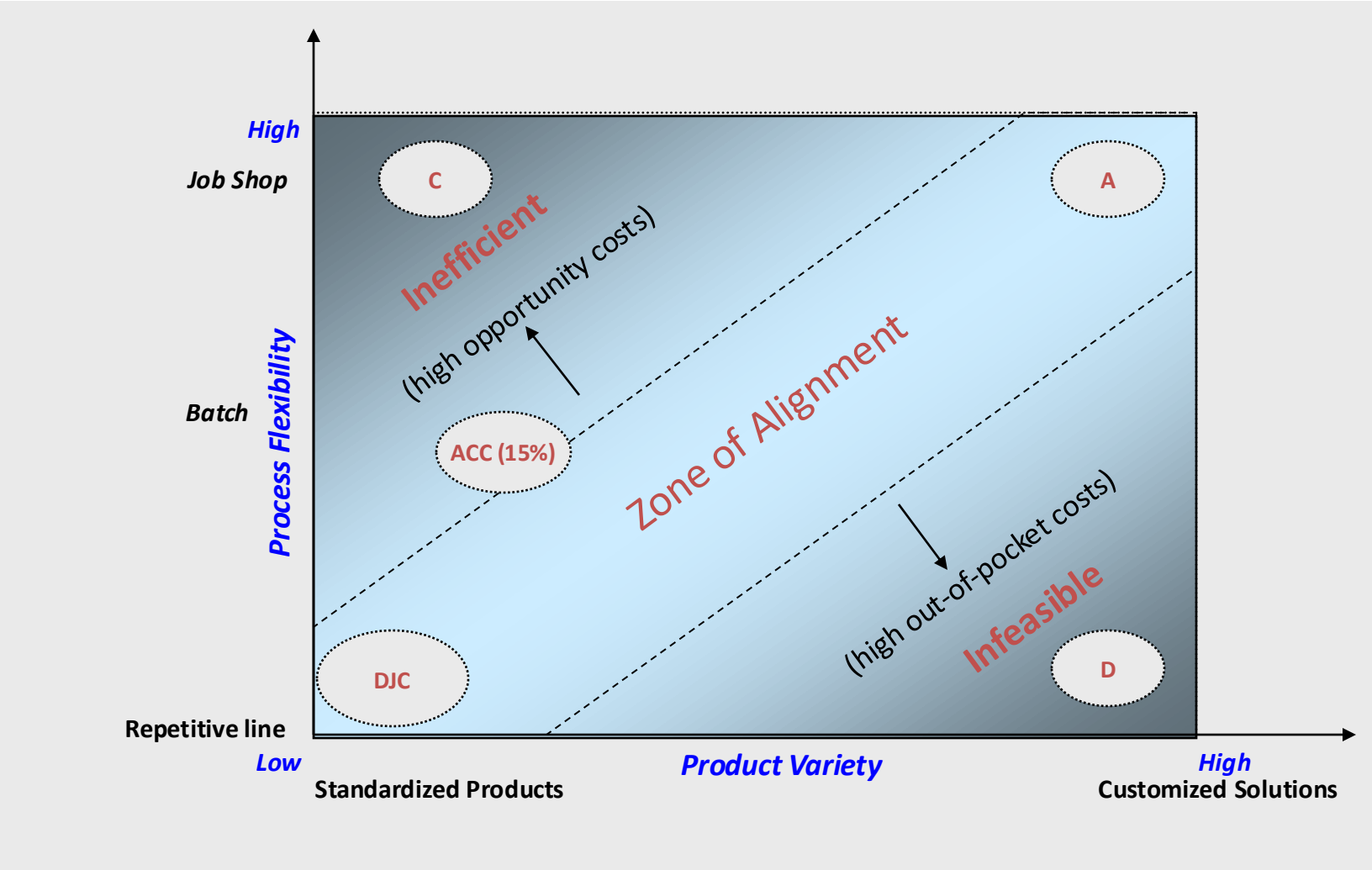
ACC: Disconnected line flow



- Plant divided into 5 large functional areas
- Each area produces all products types
- sufficient WIP buffers between different areas to support and allow batch production

How does this support product variety?

Accessing Alignment



Case Learning Points

- Assessing competitive operations strategies:
  - ACC and DJC have chosen different competitive strategies, resources, and processes
    - DJC offers a standard product portfolio (sells a "manufacturing capability") and is formidable learner
    - ACC offers design & mfg customization (sell "customer solutions") but is not-aligned for a large part of their business
  - Tools: framework elements + alignment (product-process matrix)
- Valuing an operations' differentiation strategy must correct for differences in
  1. Location
  2. Volume and other deviations from planned operating conditions
  3. Strategic positioning
  4. The remaining cost differential is attributable to operational efficiency
     - This break-down quantifies the trade-offs between competing on cost vs. innovation/flexibility
- Designing strategic defense
  1. On the operations side: definitely implement OE improvement activities to eliminate ΔC_{OE} and safeguard own custom segment
     - The magnitude of the different items dictate the priority in implementation
     - Consider "standard segment": is current process best fit, implement separate tailored process, or exit it?
  2. On the value side: how can we improve ΔValue?
  3. On the competitive side: Thwart DJC entry: prevent them from gathering scale …
- Insights: Benchmarking must correct for inherent differences between apples and oranges. Sometimes companies that look like competitors may not be competitors
  - Critical to sort out differences due to strategy and those due to pure inefficiency. Without controlling for strategy differences, the situation may look worse or better than it actually is.
  - Use benchmarking to target those capabilities that need improvement to better carry out strategy